
Reshaping Reasoning in LLMs: A Theoretical Analysis of RL Training Dynamics through Pattern Selection

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Abstract

1 Reinforcement learning (RL) has driven remarkable gains in the reasoning capa-
2 bilities of language models, yet the training dynamics behind these gains have
3 remained opaque. In this work, we resolve this gap, delivering a unified account of
4 the RL training process that rests on both systematic empirical analysis and rigorous
5 theoretical modeling. Through fine-grained reasoning-pattern-level and token-level
6 analysis across the entire RL training process, we establish a clear and consistent
7 picture: the success rates of individual reasoning patterns stay essentially fixed
8 during training, and RL improves performance not by learning new reasoning but
9 by optimizing a strikingly sparse subset of critical tokens, which in turn reshapes
10 the distribution over reasoning patterns. Building directly on these observations,
11 we develop a theoretical framework that pins down the training dynamics of RL
12 under two canonical reward types: verifiable reward (RLVR) and the model’s
13 internal feedback (RLIF). For RLVR, we prove that the dynamics fall into exactly
14 two regimes—one in which the model converges rapidly to the optimal reasoning
15 strategy, and one in which optimization stalls—and we identify the base model’s
16 reasoning quality as the decisive factor governing which regime holds. For RLIF,
17 we prove why internal rewards improve performance early in training and why
18 continued training must eventually degrade it. Extensive experiments confirm every
19 prediction of our theory, together advancing both the theoretical understanding and
20 the practical deployment of RL for language model enhancement.

21 1 Introduction

22 State-of-the-art reasoning models such as Gemini2.5 [8], Qwen3 [40], and DeepSeek-R1 [14] now
23 achieve exceptional performance on complex logical tasks spanning mathematics [30, 44, 41] and
24 programming [49, 40, 8]. Reinforcement learning (RL) is the key technique behind this success,
25 repeatedly proving able to elevate model capabilities to a new level.

26 The success of RL has triggered intense research into its underlying mechanisms for LLMs. By
27 comparing pass@k performance, Yue et al. [42] show that models post-trained with RL struggle to
28 surpass base models, indicating that RL does not elicit fundamentally new reasoning patterns. From
29 an entropy perspective, Cui et al. [9], Zhang et al. [45] prove that RL-based methods reduce policy
30 entropy, and Cui et al. [9] further establish an empirical link between model performance and policy
31 entropy. Wang et al. [33], Huan et al. [17] demonstrate that RL primarily optimizes a sparse subset of
32 critical tokens. Valuable as they are, these efforts each capture only one facet of the phenomenon, and
33 the training dynamics that actually drive RL remain unexplained—both empirically and theoretically.
34 We close this gap.

35 To pin down the RL training process, we begin with a systematic reasoning-pattern-level and token-
36 level analysis across RL training. Focusing squarely on the training dynamics, we examine both
37 the ranking shifts across training and, using complementary LLM-based and rule-based methods,

extract and classify the reasoning patterns in the models’ responses, tracking their success rates and distributions throughout training. Our experiments deliver markedly clearer and more decisive evidence than prior work [17, 42]: RL optimizes a sparse subset of critical tokens and thereby *reshapes reasoning pattern distributions* to drive model performance. We further establish that the intrinsic success rate of individual patterns stays essentially stable. These insights directly motivate the mathematical framework we build to explain the RL training process.

Grounded in these empirical findings, we develop a theoretical framework that conceptualizes reasoning as a two-stage question-reason-answer process $q \rightarrow r \rightarrow a$: (1) reasoning pattern selection given the question, i.e., $\pi(r|q)$, and (2) answer generation given the chosen pattern, i.e., $\pi(a|r, q)$. Within this framework we characterize the training dynamics of RL in LLMs under two canonical reward types: verifiable reward (RLVR) [14, 30] and the model’s internal feedback (RLIF) [47, 4]. For RLVR, we prove that it converges to the reasoning pattern r^* with the highest success rate, exactly matching our empirical observations. We further show that the dynamics split into precisely two regimes: models with strong initial reasoning quality converge rapidly to optimal patterns, whereas weaker models are forced through an entanglement stage that provably stalls optimization. For RLIF, we prove both why internal rewards improve performance early in training and why such methods must ultimately converge to a state *worse* than the base model—again matching what we observe empirically. Additional experiments confirm every element of our theoretical analysis.

The main contributions of this paper are highlighted as follows:

- We conduct a systematic reasoning-pattern-level and token-level analysis across the RL training process. By examining next-token-prediction ranking shifts together with LLM-based and rule-based reasoning pattern analysis, we deliver substantially clearer and more decisive evidence on RL training dynamics than prior work [17, 42]. Our experiments establish that RL optimizes a sparse subset of critical tokens and thereby *reshapes reasoning pattern distributions* to drive model performance, while the intrinsic success rate of individual patterns stays essentially stable throughout training.
- We develop a formal two-stage mathematical framework that models reasoning as $q \rightarrow r \rightarrow a$ (question-reason-answer) and use it to characterize the training dynamics of two canonical RL-based approaches. For RLVR, we prove convergence to the reasoning pattern with the highest success rate and show the dynamics split into exactly two regimes: rapid convergence for strong base models versus a provably stalled entanglement stage for weaker models. For RLIF, we prove both the performance improvement at early training stages and why such methods must ultimately converge to worse performance than the base model.
- We validate the full theoretical analysis through additional case studies. Together, these results close the gap between empirical observation and theoretical understanding of RL training dynamics in LLMs, and yield concrete, actionable guidance for improving RL-based LLM post-training.

2 Related Works

Reinforcement Learning for LLMs Reinforcement learning has driven remarkable gains in large language models (LLMs), both in aligning models with human preferences [25, 48] and in solving complex mathematical and programming tasks [30, 20, 22]. A central component of RL is the reward model, which traditionally depends on extensively trained, human-annotated datasets [26, 25, 2]. The paradigm has since shifted decisively toward more easily obtainable and verifiable rewards, as in reinforcement learning with verifiable rewards (RLVR) [30, 40, 14] and reinforcement learning with internal feedback (RLIF) [47, 4]. This success has, in turn, spurred a wave of new RL algorithms, including GRPO [30], DAPO [41], and VAPO [43].

Theoretical Analyses and Mechanism Interpretation of RL for LLMs. The success of RL in LLMs has triggered extensive research into theoretical analyses and mechanism interpretation. Initial efforts focused mainly on high-level aspects of RL, such as how to design better rewards [28, 18, 38] and how to develop novel training algorithms [37, 11, 21]. As RL paradigms shifted toward simpler reward structures like RLVR and RLIF, researchers sought the fundamental reasons behind their effectiveness from various angles. Shao et al. [29] showed that spurious rewards enhance reasoning by unlocking latent skills from pretraining, while Agarwal et al. [4] attributed performance improvements to entropy minimization. Gandhi et al. [13] identified specific cognitive behaviors,

such as backtracking, that improve reasoning. Zhao et al. [46] demonstrated that RL fine-tuning amplifies pre-trained behaviors and converges toward dominant output formats, a phenomenon further analyzed by Wang et al. [34], Cui et al. [10] through the lens of entropy collapse. Each of these accounts explains a single symptom in isolation. In contrast, we analyze RL training dynamics directly from the perspective of reasoning pattern selection, and, to our knowledge, are the first to back this perspective with a complete mathematical framework and rigorous theoretical analysis.

3 Preliminaries

Reinforcement Learning for LLMs Let π_θ be a language model with parameters θ , which serves as the policy to be optimized. Given an input question $\mathbf{x} = (x_0, x_1, \dots, x_n)$, the policy π_θ generates an answer $\mathbf{y} = (y_0, y_1, \dots, y_m)$. The optimization objective can be formulated as:

$$\phi_{\text{RL}}(\theta) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}, \mathbf{y} \sim \pi_\theta(\cdot | \mathbf{x})} [r_\phi(\mathbf{x}, \mathbf{y})] - \beta \mathbb{D}_{\text{KL}}[\pi_\theta(\mathbf{y} | \mathbf{x}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})], \quad (3.1)$$

where π_{ref} is the base reference policy, and β is a hyperparameter that controls the KL divergence to prevent excessive deviation between π_θ and π_{ref} .

The reward function $r_\phi(\mathbf{x}, \mathbf{y})$ can be implemented in various formats, such as a trained reward model [25] or a rule-based scoring function [30]. In this paper, we focus on verifiable reward (RLVR) [14, 30] and the model’s internal feedback (RLIF) [47, 4].

In RLVR, the reward function $r_\phi(\mathbf{x}, \mathbf{y})$ directly evaluates whether the answer $\mathbf{y}$ matches the correct answer to question $\mathbf{x}$. A typical reward function is defined as:

$$r_\phi(\mathbf{x}, \mathbf{y}) = \begin{cases} 1 & \text{if } \mathbf{y} = \text{the ground truth of } \mathbf{x}, \\ 0 & \text{otherwise.} \end{cases} \quad (3.2)$$

Unlike RLVR, RLIF [47, 4] leverages rewards based solely on intrinsic model-derived signals. Following Zhao et al. [47], we consider the RLIF reward as the negative average KL divergence between a uniform distribution U over the vocabulary $\mathcal{V}$ and the model’s next-token distribution

$$r_\phi(\mathbf{x}, \mathbf{y}) = \frac{1}{|\mathbf{y}|} \sum_{i=1}^{|\mathbf{y}|} \mathbb{D}_{\text{KL}}(U \parallel \pi_\theta(\cdot | \mathbf{x}, \mathbf{y}_{<i})) = -\frac{1}{|\mathbf{y}| \cdot |\mathcal{V}|} \sum_{i=1}^{|\mathbf{y}|} \sum_{j=1}^{|\mathcal{V}|} \log(|\mathcal{V}| \cdot \pi_\theta(\cdot | \mathbf{x}, \mathbf{y}_{<i})). \quad (3.3)$$

In this paper, we characterize the training dynamics of both RLIF and RLVR through combined empirical and theoretical analysis.

4 Experimental Exploration for RL Training

To pin down the RL training process, we conduct systematic experiments and analysis across RL training with different rewards. We begin with a high-level overview of the training procedure, which shows unambiguously that RLVR yields steady improvements whereas RLIF is unstable and can outright degrade performance. To dissect these dynamics, we perform a reasoning-pattern-level analysis, which reveals that RLVR’s success stems precisely from its tendency to adopt reasoning patterns with higher success rates, while RLIF fails to specialize in these patterns at all. Finally, a token-level analysis exposes the underlying mechanism behind these pattern dynamics: they are driven by rank changes in a strikingly small fraction of tokens. Compared with prior work [17, 42], our experiments provide far clearer and more decisive evidence of these RL training dynamics, establishing that RL optimizes a sparse subset of critical tokens and thereby *reshapes reasoning pattern distributions* to drive model performance.

4.1 Training Procedure Overview for RLVR and RLIF

We first analyze the complete training procedures for both RLVR and RLIF. For a controlled comparison, we take Qwen2.5-3B [39] as the base model and train it on the MATH dataset [16], holding all other settings identical across the two algorithms.

As Figure 1a makes clear, the two methods follow sharply different performance trajectories. RLVR yields a stable training process in which performance improves continuously and converges smoothly.

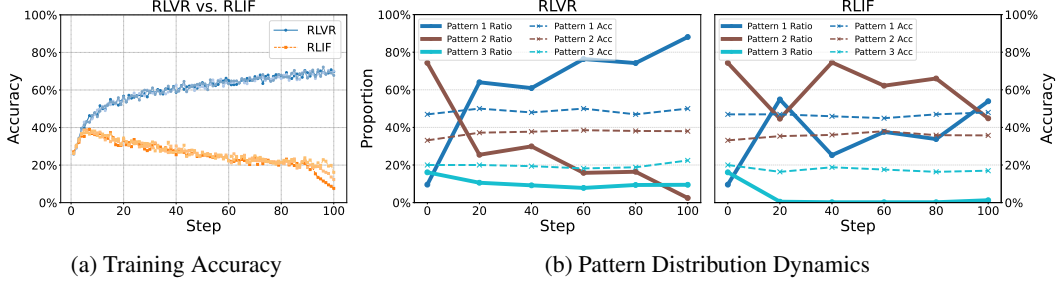


Figure 1: **(a)** The training procedures of RLVR and RLIF on the MATH dataset. The finding is unambiguous: RLVR delivers stable gains, whereas RLIF drives performance up and then back down. We repeat each training paradigm across multiple random seeds. **(b)** The reasoning-pattern-level comparison. The solid line gives the proportion of a given pattern among all patterns in the responses (left vertical axis); the dotted line gives the accuracy of that pattern (right vertical axis). Under RLVR, the model steadily shifts toward higher-accuracy patterns and abandons lower-accuracy ones, whereas RLIF exhibits unstable dynamics. For ease of observation, we sort and name the patterns from high to low according to their accuracy.

131 RLIF behaves in the opposite manner: it improves performance early on, but continued training
 132 drives performance down, in some runs below that of the original base model. This contrast directly
 133 motivates the systematic, fine-grained empirical analysis we carry out next to uncover the underlying
 134 RL training dynamics.

135 4.2 Reasoning-pattern Level Analysis

136 To explain the performance gap between RLVR and RLIF, we conduct a fine-grained analysis at the
 137 reasoning-pattern level. To define and extract these patterns, we collect responses from the base model
 138 and employ GPT-4o [19] to group them into distinct categories by keywords and logical structure,
 139 generating a description for each pattern. We then use these classifications to track the reasoning
 140 pattern distribution and its corresponding success rates throughout the RL training procedure.

141 Specifically, we analyze the dynamics of reasoning patterns and their accuracy during both RLVR
 142 and RLIF training. Figure 1b presents our analysis on responses to Number Theory tasks from the
 143 MATH dataset [16], with additional results across models and datasets in Appendix A. The analysis
 144 yields three clear findings:

- 145 • RLVR-trained models consistently shift toward reasoning patterns with higher success rates, which
 146 is exactly what drives the steady improvement in overall accuracy under RLVR.
- 147 • RLIF, by contrast, never specializes: its reasoning pattern distribution remains unstable throughout
 148 training and fails to lock onto the more effective patterns.
- 149 • The success rate of any individual reasoning pattern stays stable throughout training for both
 150 methods.

151 This analysis establishes that the performance gap between RLVR and RLIF is fully accounted for
 152 by their distinct reasoning pattern dynamics. We therefore turn to the mechanism that drives these
 153 changes during RL training.

154 4.3 Token-level Analysis

155 To expose the mechanism behind the reasoning
 156 pattern dynamics observed during RL training,
 157 and building on Huan et al. [17], Yue et al. [42],
 158 we conduct a token-level analysis. We sample re-
 159 sponses from the base model for questions from
 160 the GSM8K [6], MATH [16], and AIME [7]
 161 datasets. For these identical responses, we then
 162 examine the corresponding token ranks in both

Table 1: The fraction of tokens whose rank changes after RL—consistently small across tasks and steps.

Task	Method	Step20	Step40	Step60	Step80	Step100
GSM8K	RLVR	5.2%	6.0%	6.6%	7.1%	7.3%
	RLIF	6.8%	8.0%	8.9%	9.5%	10.1%
MATH	RLVR	5.3%	5.8%	6.1%	6.5%	6.6%
	RLIF	6.1%	7.0%	7.7%	8.3%	8.8%
AIME24	RLVR	5.1%	5.7%	5.9%	6.1%	6.3%
	RLIF	5.6%	6.5%	7.4%	8.2%	8.6%

the base and the RL-enhanced models, where a token’s rank is the position of its probability among all vocabulary tokens given the preceding context. As Table 1 shows, individual token ranks remain overwhelmingly stable, with fewer than 10% of tokens shifting rank in most cases. This pins down the mechanism precisely: RL selectively edits the probabilities of a sparse set of critical tokens and leaves the vast majority of the reasoning process untouched.

Taken together, our experiments deliver clean and decisive evidence for RL training dynamics: *RL optimizes a sparse subset of critical tokens, thereby reshaping reasoning pattern distributions to drive model performance, while the intrinsic success rate of individual patterns stays essentially stable during training.*

These empirical findings do more than deepen our understanding of RL training dynamics: they supply the exact structure on which we build the mathematical framework for the theoretical analysis that follows.

5 Theoretical Characterization of RL Training Dynamics

We now move beyond empirical observation to a rigorous theoretical account of the RL training process. We first build a theoretical framework directly from our experimental insights, abstracting the model’s reasoning procedure as a two-step process: first selecting a reasoning pattern, then deducing the answer, with RL acting specifically on the first step. Within this framework we prove the behavior of RL under different rewards. For RLVR, we characterize its training dynamics and show that two distinct optimization regimes arise. For RLIF, we prove both why internal rewards can improve model performance and why such methods must ultimately yield models worse than the base model.

5.1 A Theoretical Framework for Reasoning Models

Abstract Reasoning Process. We formalize the reasoning process as follows: given a question q , the model (1) selects a reasoning pattern r from candidate patterns $\mathcal{R} = \{r_1, r_2, \dots\}$ and (2) generates a final answer $a \in \mathcal{A} = \{a_1, a_2, \dots\}$ accordingly. The model selects reasoning pattern r_i with probability $p(r_i|q)$, and each reasoning pattern has a distinct success rate. Within this framework, we can reformulate the RLIF optimization objective (Eq. 3.3) as follows¹:

$$r_\phi(\{r, a\}, q) := - \left(\frac{1}{|\mathcal{R}|} \sum_{r_i \in \mathcal{R}} \log(|\mathcal{R}| \cdot \pi_\theta(r_i|q)) + \frac{1}{|\mathcal{A}|} \sum_{a_j \in \mathcal{A}} \log(|\mathcal{A}| \cdot \pi_\theta(a_j|q, r)) \right) \quad (5.1)$$

Policy Parameterization. Let $\mathcal{V} = \{q\} \cup \mathcal{R} \cup \mathcal{A}$ denote the vocabulary set and $\mathcal{Y} = (r, a)_{r \in \mathcal{R}, a \in \mathcal{A}}$ represent the set of output sequences. Given a question q , the language model produces a distribution over output sequences $y \in \mathcal{Y}$ autoregressively:

$$(\text{general policy}) \quad \pi_\theta(y|x) = \prod_{l=1}^{|y|} \pi_\theta(y_l|x, y_{<l}) = \prod_{l=1}^{|y|} \text{softmax}(f_\theta(x, y_{<l}))_{y_l}, \quad (5.2)$$

where $f_\theta : \mathcal{V} \rightarrow \mathbb{R}^{|\mathcal{V}|}$ is a function parameterized by θ , and the model predicts the l -th token based on the previous context $y_{<l}$. The next token follows the distribution $\text{softmax}(z)_v := \exp(z_v) / \sum_{v' \in \mathcal{V}} \exp(z_{v'})$ for $z = f_\theta(x, y_{<l})$.

Due to the complexity of practical models, establishing optimization guarantees for understanding RL training dynamics has proven very challenging [3, 23]. Following previous works [27, 24, 10, 45], we consider a *tabular policy* parameterization, which can be viewed as a special case of Eq. 5.2 where each output is assigned its own trainable logit for the corresponding last token, i.e., for $\theta \in \mathbb{R}^{|\mathcal{V}| \times |\mathcal{V}|}$:

$$(\text{tabular policy}) \quad \pi_\theta(y_l|x, y_{<l}) = \pi_\theta(y_l|y_{l-1}) = \text{softmax}(\theta_{:,y_{l-1}})_{y_l}, \quad (5.3)$$

where $\theta_{:,y_{l-1}} \in \mathbb{R}^{|\mathcal{V}|}$ is the column of θ corresponding to y_{l-1} .

¹For simplicity, we omit the normalizing coefficient $1/|y|$ from Eq. 3.3, which can be treated as a constant $1/2$ in our framework.

202 **Optimization for RL.** For our analysis of training dynamics, we consider the tabular policy from
 203 Eq.5.3 with the optimization objective $\phi_{\text{RL}}(\theta)$ in Eq.3.1. We analyze the policy gradient in the small
 204 learning rate limit using gradient flow:

$$\frac{d}{dt}\theta(t) = \nabla\phi_{\text{RL}}(\theta(t)), \quad t \geq 0 \quad (5.4)$$

205 where $\theta(t)$ represents the parameters of the policy $\pi_{\theta(t)}$ at training time t , initialized with $\theta(0) = \theta_{\text{ref}}$.

206 Our experiments established that success rates for individual reasoning patterns remain stable during
 207 training. This stability follows directly from the model’s architectural constraints: optimizing the
 208 mapping from questions to reasoning patterns is far easier than optimizing the path from reasoning
 209 patterns to final answers. We formalize this observation as the following assumption:

210 **Assumption 5.1.** *The success rate for each reasoning pattern $\mathbf{r}_i \in \mathcal{R}$ to provide the correct answer*
 211 *$\mathbf{r}^*$, defined as $\pi_{\theta}(\mathbf{a}^*|\mathbf{q}, \mathbf{r}_i)$, remains constant during training.*

212 In the remaining part of this section, we adopt Assumption 5.1 and use $p^*(\mathbf{r}) = \pi_{\theta}(\mathbf{a}^*|\mathbf{q}, \mathbf{r})$ to
 213 denote the fixed success rate for the given pattern $\mathbf{r}$. We first derive the optimal policy for the RL
 214 objective (Eq.3.1) under the general autoregressive policy (Eq.5.2). We then analyze RLVR and RLIF
 215 training dynamics using the tabular policy (Eq. 5.3).

216 5.2 The Optimal Policy for RL

217 Our empirical results show that RLVR improves performance steadily by incentivizing reasoning
 218 patterns with higher success rates, whereas RLIF yields only unstable improvement. We now explain
 219 both behaviors from first principles, beginning with a precise characterization of the optimal policy
 220 for the RLVR and RLIF objectives:

221 **Proposition 5.2.** *Suppose we maximize the RL objective (Eq 3.1) using a general autoregressive*
 222 *policy (Eq 5.2) and Assumption 5.1 holds. Then, the optimal policy satisfies:*

$$\pi_{\text{opt}}(\mathbf{r}|\mathbf{q}) = \frac{1}{Z} \exp\left(\frac{1}{\beta}R(\mathbf{r})\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q}) \text{ for all } \mathbf{r} \in \mathcal{R}, \quad (5.5)$$

223 where $Z = \sum_{\mathbf{r} \in \mathcal{R}} \exp\left(\frac{1}{\beta}R(\mathbf{r})\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q})$ is the normalizing coefficient, and $R(\mathbf{r})$ denotes the
 224 reasoning path reward. Specifically:

- 225 • For RLVR, $R(\mathbf{r})$ equals the success rate of reasoning pattern $\mathbf{r}$, i.e., $R_{\text{RLVR}}(\mathbf{r}) = p^*(\mathbf{r})$.
- 226 • for RLIF, $R(\mathbf{r})$ equals to the confidence for the reasoning pattern $\mathbf{r}$ for final answer distribution,
 227 i.e., $R_{\text{RLIF}}(\mathbf{r}) = -\frac{1}{|\mathcal{A}|} \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta}(\mathbf{a}|\mathbf{q}, \mathbf{r}))$.

228 Proposition 5.2 pins down the optimal solutions for RLVR and RLIF, and the contrast between them
 229 is decisive. Assuming π_{opt} resides within the general autoregressive policy parameterized space
 230 (which always holds for both LLMs and the tabular policies discussed later), the RLVR-optimized
 231 model increases the probability of a reasoning pattern $\mathbf{r}$ exactly when its probability of deducing
 232 the correct answer $p^*(\mathbf{r})$ is high enough that $\exp(\frac{1}{\beta}p^*(\mathbf{r}))/Z > 1$. For RLIF, however, $R_{\text{RLIF}}(\mathbf{r})$
 233 depends only on the confidence of the final answer distribution and therefore cannot distinguish the
 234 correct answer $\mathbf{a}^*$ from incorrect alternatives. The consequence is clear-cut: the optimal policy under
 235 RLVR consistently improves performance, whereas the optimal policy under RLIF carries no such
 236 guarantee and may fall below the base model.

237 Moreover, since β is typically small (e.g., 0.001) in practice [26, 5, 44], as $\beta \rightarrow 0$, we have:

$$\pi_{\text{opt}}(\mathbf{r}|\mathbf{q}) = \lim_{\beta \rightarrow 0} \frac{1}{Z} \exp\left(\frac{1}{\beta}R(\mathbf{r})\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q}) = \begin{cases} 1 & \mathbf{r} = \arg \max_{\mathbf{r}} R(\mathbf{r}) \\ 0 & \text{otherwise} \end{cases}, \quad (5.6)$$

238 In this limit the policy converges to a deterministic strategy that always selects the reasoning pattern
 239 $\mathbf{r}$ with the highest $R(\mathbf{r})$, regardless of how the initialized reference model chooses reasoning patterns.
 240 The optimal solution alone, however, does not reveal the *path* to it: because of the non-convexity
 241 of the optimization landscape, we still need to explain how RLVR reliably reaches high-reward
 242 patterns and why RLIF exhibits early performance improvements. We resolve both questions next by
 243 analyzing the training dynamics of the two methods under a tabular policy (Eq. 5.3).

244 5.3 Training Dynamic Analysis for RLVR

245 Our experiments demonstrate that RLVR improvements hinge on the capacity of the base model. We
 246 focus on the probability of the *optimal reasoning pattern*, i.e., the pattern $\mathbf{r}^*$ with the highest success
 247 rate for reaching the correct answer $\mathbf{a}^*$, $\mathbf{r}^* = \arg \max_{\mathbf{r}} p^*(\mathbf{r})$. Analyzing the gradient flow, we prove
 248 that the training dynamics of RLVR fall into exactly two regimes. In the first, the probability of the
 249 optimal reasoning pattern $\pi_{\theta}(\mathbf{r}^*|\mathbf{q})$ increases monotonically until it converges to 1. In the second,
 250 the model is first trapped in an entanglement stage, where a suboptimal reasoning pattern $\mathbf{r}' \neq \mathbf{r}^*$
 251 obstructs the optimization of the optimal pattern; only after this stage does the model transition to the
 252 dynamics of the first regime and converge to $\mathbf{r}^*$.

253 **Theorem 5.3** (Regime 1: Sufficient Condition for Efficient Convergence). *Consider the RLVR (Eq*
 254 *3.2) with $\beta = 0$ for optimizing objective Eq 3.1, using a tabular policy (Eq 5.3) with Assumption 5.1*
 255 *holds. Let $\mathbf{r}^*$ be the optimal reasoning pattern, if the overall accuracy of the initialized model $\pi_{\theta_{\text{ref}}}$,*
 256 *defined as $\text{ACC}_{\theta_{\text{ref}}} = \sum_{\mathbf{r} \in \mathcal{R}} \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q})p^*(\mathbf{r})$, satisfies:*

$$(Regime 1) \quad \text{ACC}_{\theta_{\text{ref}}} > p^*(\mathbf{r}) \text{ for all } \mathbf{r} \in \mathcal{R}, \mathbf{r} \neq \mathbf{r}^*, \quad (5.7)$$

257 *then for any $\epsilon > 0$, there exists $T_1 = \mathcal{O}(\frac{1}{\epsilon})$ such that for $t > T_1$, we have $1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) < \epsilon$.*

258 Theorem 5.3 covers the case in which the base reference model is strong enough that its overall
 259 accuracy exceeds the success rate of every non-optimal reasoning pattern (Eq 5.7). Here RLVR
 260 provably guides the model to the optimal reasoning pattern at rate $\mathcal{O}(1/\epsilon)$, attaining an overall
 261 accuracy close to the success rate of that optimal pattern.

262 RLVR optimization, however, is known to be difficult in some settings [44, 36], and this difficulty
 263 arises precisely when the base reference model is weaker. Our second regime captures exactly this
 264 case: the model is first trapped in an entanglement stage in which the *suboptimal reasoning pattern*
 265 *with the second-highest success rate*, $\mathbf{r}' = \arg \max_{\mathbf{r}, \mathbf{r} \neq \mathbf{r}^*} p^*(\mathbf{r})$, slows down the optimization of the
 266 optimal reasoning pattern:

267 **Theorem 5.4** (Regime 2: Slow Convergence for optimal reasoning pattern). *Consider the RLVR (Eq*
 268 *3.2) with $\beta = 0$ for optimizing objective Eq 3.1, using a tabular policy (Eq 5.3) with Assumption 5.1*
 269 *holds. Let $\mathbf{r}^*$ and $\mathbf{r}'$ be the optimal and second optimal reasoning patterns, if the overall accuracy of*
 270 *the initialized model satisfies:*

$$(Regime 2) \quad p^*(\mathbf{r}') > \text{ACC}_{\theta_{\text{ref}}} > p^*(\mathbf{r}) \text{ for all } \mathbf{r} \in \mathcal{R}/\{\mathbf{r}^*, \mathbf{r}'\}, \quad (5.8)$$

271 *then there exists:*

$$T_0 = \frac{1}{2 - 2\pi_{\theta_{\text{ref}}}(\mathbf{r}'|\mathbf{q})} \left((C_1 \cdot \gamma_{\pi_{\text{ref}}})^{2C_2 \cdot \gamma_{\pi_{\text{ref}}}} - 1 \right), \text{ where } \gamma_{\pi_{\text{ref}}} := \sum_{\mathbf{r} \in \mathcal{R}/\{\mathbf{r}'\}} \frac{\pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q})}{\pi_{\theta_{\text{ref}}}(\mathbf{r}^*|\mathbf{q})} \quad (5.9)$$

272 *with constants C_1, C_2 depending on the success rates of reasoning patterns, such that for we can*
 273 *guarantee the model transform from regime 2 (Eq 5.8) to regime 1 (Eq 5.7), i.e., $\text{ACC}_{\theta(t)} >$*
 274 *$p^*(\mathbf{r}), \forall \mathbf{r} \in \mathcal{R}, \mathbf{r} \neq \mathbf{r}^*$ for $t \geq T_0$.*

275 Theorem 5.4 treats the regime in which only the optimal and suboptimal reasoning patterns have
 276 success rates exceeding the overall accuracy. We adopt this case for analytical clarity; the same
 277 argument extends to the more general setting in which *at least* two reasoning patterns exceed the
 278 average success rate, at the cost of a heavier analysis.

279 The decisive insight of Theorem 5.4 is that RLVR may need T_0 time steps before the overall success
 280 rate overtakes that of the suboptimal reasoning pattern—that is, before it even reaches the favorable
 281 regime of Theorem 5.3. Whereas the convergence time T_1 in Theorem 5.3 is polynomial in $1/\epsilon$, the
 282 time T_0 in Theorem 5.4 grows super-exponentially in $\gamma_{\pi_{\text{ref}}}$ —the ratio between the total success rate
 283 through $\mathcal{R}/\{\mathbf{r}'\}$ and the success rate through $\mathbf{r}^*$. When the base model assigns a very small selection
 284 probability to the optimal reasoning pattern, $\gamma_{\pi_{\text{ref}}}$ becomes large and T_0 prohibitive. This is exactly
 285 the entanglement stage: the model can require an extraordinarily long training period before it selects
 286 the optimal reasoning pattern with appreciable probability.

287 Sec 6 provides additional experiments and case studies that confirm both regimes.

5.4 Theoretical Explanation for RLIF

Proposition 5.2 already shows that RLIF, depending only on the confidence of the final answer distribution, cannot distinguish the correct answer $\mathbf{a}^*$ from incorrect alternatives—so RLIF must eventually degrade performance relative to the base model, exactly as we observe empirically. Yet prior studies report that RLIF can improve model performance without external rewards Agarwal et al. [4], Zhao et al. [47]. We reconcile these two facts, proving why RLIF improves the performance of a well-trained LLM at the initial training stage even though it must fail later.

First, we consider a well-trained LLM satisfying the following assumption:

Assumption 5.5. *For the base model, the correct answer $\mathbf{a}^*$ has the highest probability across all possible answers, i.e., $\arg \max_{\mathbf{a} \in \mathcal{A}} \sum_{\mathbf{r} \in \mathcal{R}} \pi_{\text{base}}(\mathbf{a}|\mathbf{q}, \mathbf{r}) \pi_{\text{base}}(\mathbf{r}|\mathbf{q}) = \mathbf{a}^*$.*

Because majority voting [35] is a proven route to better performance, a well-trained LLM assigns a higher probability to the correct answer $\mathbf{a}^*$ than to any incorrect answer, so Assumption 5.5 holds for modern LLMs. Under it, we consider the case $|\mathcal{A}| = 2$, with each reasoning path’s success rate following a uniform distribution and high-entropy reasoning pattern selection². Analyzing the overall accuracy dynamics at $t = 0$ then yields the following theorem:

Theorem 5.6 (RLIF Increases Overall Accuracy at Initialization). *Consider the RLIF (Eq 5.1) with $\beta = 0$ for optimizing objective Eq 3.1, using a tabular policy (Eq 5.3) with Assumption 5.1 and 5.5 hold, $\pi_{\theta_{\text{base}}}(\mathbf{r}|\mathbf{q}) = \frac{1}{|\mathcal{R}|}$ for all $\mathbf{r} \in \mathcal{R}$, $|\mathcal{A}| = 2$ and success rate for each reasoning path follows a uniform distribution $p^*(\mathbf{r}) \sim U[0, 1]$. Then when $|\mathcal{R}| \rightarrow +\infty$, the following holds:*

1. *The accuracy derivative at $t = 0$ is positive with probability 1, i.e. $P(\frac{d}{dt} \text{ACC}_{\theta(t)}|_{t=0} > 0) = 1$.*
2. *With probability $p = 0.5$, that $\arg \max_{\mathbf{r}} R_{\text{RLIF}}(\mathbf{r}) = \arg \min_{\mathbf{r}} p^*(\mathbf{r})$.*

The first result in Theorem 5.6 shows that overall accuracy increases at initialization, giving a rigorous explanation for RLIF’s early performance gains. As training continues, however, Proposition 5.2 forces the policy (at $\beta = 0$) to converge deterministically to the reasoning pattern $\mathbf{r}$ that maximizes $R_{\text{RLIF}}(\mathbf{r})$. The second result in Theorem 5.6 then delivers the sting: with probability 0.5, this maximizing pattern is exactly the one that *minimizes* the success rate, $\arg \max_{\mathbf{r}} R_{\text{RLIF}}(\mathbf{r}) = \arg \min_{\mathbf{r}} p^*(\mathbf{r})$ —a full 50% chance of converging to the least accurate reasoning path. RLIF therefore improves performance early (the accuracy derivative is positive) yet, with continued training, drives the model toward very-low-accuracy reasoning paths and, in the end, below the base model. This is precisely the trajectory our experiments display.

6 Case Studies and Numerical Simulations

To validate and interpret our theoretical findings, we present case studies and training simulations that put our theorems to the test. For RLVR, we exhibit the two distinct training regimes of Theorems 5.3 and 5.4 in Fig. 2 (upper). For RLIF, we run multiple training simulations in which each reasoning path’s success rate follows a uniform distribution, confirming our analysis of RLIF’s behavior both during the initial training steps and at convergence, as shown in Fig. 2 (bottom).

6.1 Case Studies for RLVR

In Section 5.3, we proved that RLVR’s training dynamics split into two regimes. In the first (Eq 5.7), the probability of selecting the optimal reasoning pattern $\pi_{\theta}(\mathbf{r}^*|\mathbf{q})$ increases monotonically until convergence. In the second (Eq 5.8), the model is trapped in an initial entanglement phase and needs a substantially longer training period before it transitions to the first-regime dynamics and converges to the optimal reasoning pattern $\mathbf{r}^*$. Our case studies confirm both:

- As Fig. 2 (upper left) shows, once the reference model satisfies Eq 5.7—overall accuracy exceeding the success rates of all non-optimal reasoning patterns—the model converges rapidly to the optimal reasoning pattern, exactly as Theorem 5.3 predicts.
- In the second case, corresponding to Theorem 5.4, a suboptimal reasoning pattern $\mathbf{r}'$ whose success rate exceeds the initial overall accuracy forces the model through an entanglement phase:

²Previous work [33] reveals that RL primarily optimizes high-entropy tokens. Here, we consider the highest entropy case: $\pi_{\theta_{\text{base}}}(\mathbf{r}|\mathbf{q}) = 1/|\mathcal{R}|$ for all $\mathbf{r} \in \mathcal{R}$.

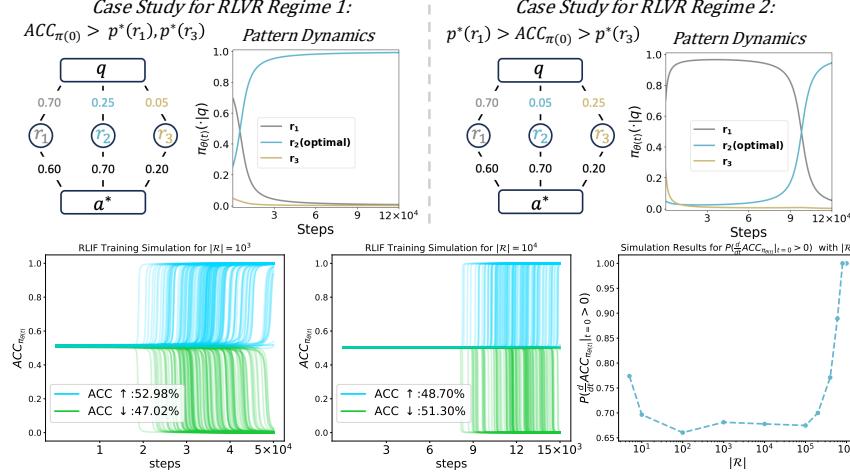


Figure 2: Case studies for RLVR (upper) and training simulations for RLIF (bottom). The upper panels confirm the two training regimes predicted for RLVR: rapid convergence (left) and the entanglement phase (right). The bottom panels report, for RLIF across varying $|\mathcal{R}|$, the probabilities of convergence to different states (left) and the probability of initial performance gains (right)—both matching our theory.

a transition period T_0 (defined in Theorem 5.4) must elapse before the model enters the rapid optimization phase of case 1. Fig. 2 (upper right) illustrates a scenario with large $\gamma_{\pi_{ref}}$ (defined in Eq 5.9): $\gamma_{\pi_{ref}} = 6$, since $(0.25 + 0.05)/0.05 = 6$. Because T_0 scales with $\gamma_{\pi_{ref}}$, it becomes prohibitively long and sharply delays convergence to the optimal reasoning pattern. The pattern dynamics in Fig. 2 display this entanglement stage unmistakably, in precise agreement with our theory.

6.2 Numerical Simulations For RLIF

In Section 5.4, we proved that under Assumption 5.5 and the base model distribution of Theorem 5.6, as $|\mathcal{R}| \rightarrow \infty$ the accuracy derivative at $t = 0$ is positive with probability 1, while the model converges to the least accurate reasoning path with probability 50%. Our simulations confirm both predictions. Fig. 2 (bottom left) reports RLIF training simulations across varying $|\mathcal{R}|$: the probability of convergence to an improved state ($ACC \uparrow$) closely matches the probability of convergence to a degraded state ($ACC \downarrow$), exactly as the second result in Theorem 5.6 requires. Fig. 2 (bottom right) examines initial-step performance and shows that for large $|\mathcal{R}|$ the probability of initial improvement approaches 1, confirming our first result in Theorem 5.6.

7 Conclusions and Limitations

This work delivers a unified account of reinforcement learning dynamics in LLMs, combining systematic empirical investigation with rigorous theory. We establish mathematical analyses for the two representative reward mechanisms RLVR and RLIF, and case studies and simulations confirm every prediction. Two natural avenues remain open for future work: sharpening the interpretability of LLM-identified reasoning patterns, and extending our framework to richer real-world reasoning scenarios and base-model regimes beyond the assumptions considered here. These are opportunities to broaden our results, not gaps in them.

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A Additional Experiments

A.1 Pattern Distribution Dynamics on RL-enhanced Qwen2.5-3B

In Section 4.2, we sampled Number Theory tasks from MATH to analyze reasoning-pattern dynamics. We reach the same conclusion on Geometry tasks (Figure 3): under RLVR the model steadily adopts the higher-accuracy reasoning patterns, whereas RLIF’s pattern selection never stabilizes. This matches both the empirical and the theoretical results of the main paper.

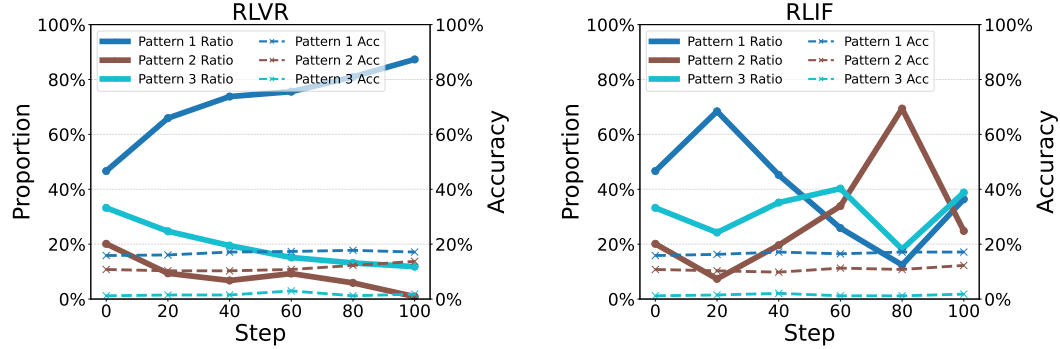


Figure 3: Dynamics of the pattern distribution and corresponding accuracy of RL-enhanced models on Geometry tasks. Under RLVR the model converges to the highest-accuracy pattern (Pattern 1), whereas RLIF remains unstable.

A.2 Qwen-2.5-7B-Instruct vs. Qwen-2.5-7B-SimpleRL-Zoo

To confirm that our findings hold across models, we run supplementary experiments that test both the experimental insights and the theoretical conclusions (Figure 4). We take Qwen-2.5-7B-Instruct [39] as the base model and compare it against Qwen-2.5-7B-SimpleRL-Zoo [44], an open-source variant enhanced with RLVR for mathematical reasoning. Our evaluation spans diverse mathematical domains—number theory, geometry, algebra, calculus, and counting and probability—using challenging problems sampled from the MATH dataset [16], and extends further to complex problems from AMC23 [1]. The results uniformly support our earlier findings: RLVR enhancement raises the frequency of high-accuracy reasoning patterns and suppresses the less effective ones. Detailed task-specific analyses appear in Appendix B.3.

Across every task, the patterns with the highest success rates consistently become more prevalent after RLVR enhancement, reinforcing our findings from Section 4 and Section 5. In tasks such as Algebra and AMC23-19 (Figure 4), the success rates of individual reasoning patterns hold steady, which both matches our earlier observations and gives direct empirical backing to Assumption 5.1 in our theoretical analysis.

A.3 Qwen-2.5-32B-Instruct vs. QwQ-32B

To verify that our theory scales, we experiment on the larger model QwQ-32B [32], which is built on Qwen-2.5-32B-Instruct [39] and substantially enhances reasoning ability through RLVR. We measure the reasoning-pattern shift between these two models on four tasks: number theory and geometry from the MATH dataset [16], a task from AIME 2024 [7], and a task from OlympiadBench [15].

The result is exactly what our theory predicts: QwQ-32B’s reasoning patterns concentrate on the most accurate patterns of Qwen-2.5-32B. We do not report QwQ’s per-pattern accuracy because we access the models through their API and cannot extract the CoT data needed to have the base model continue generating answers, so the accuracy of a reasoning pattern (as opposed to the model overall) is not recoverable here; see “Accuracy Analysis” in Appendix B.2.

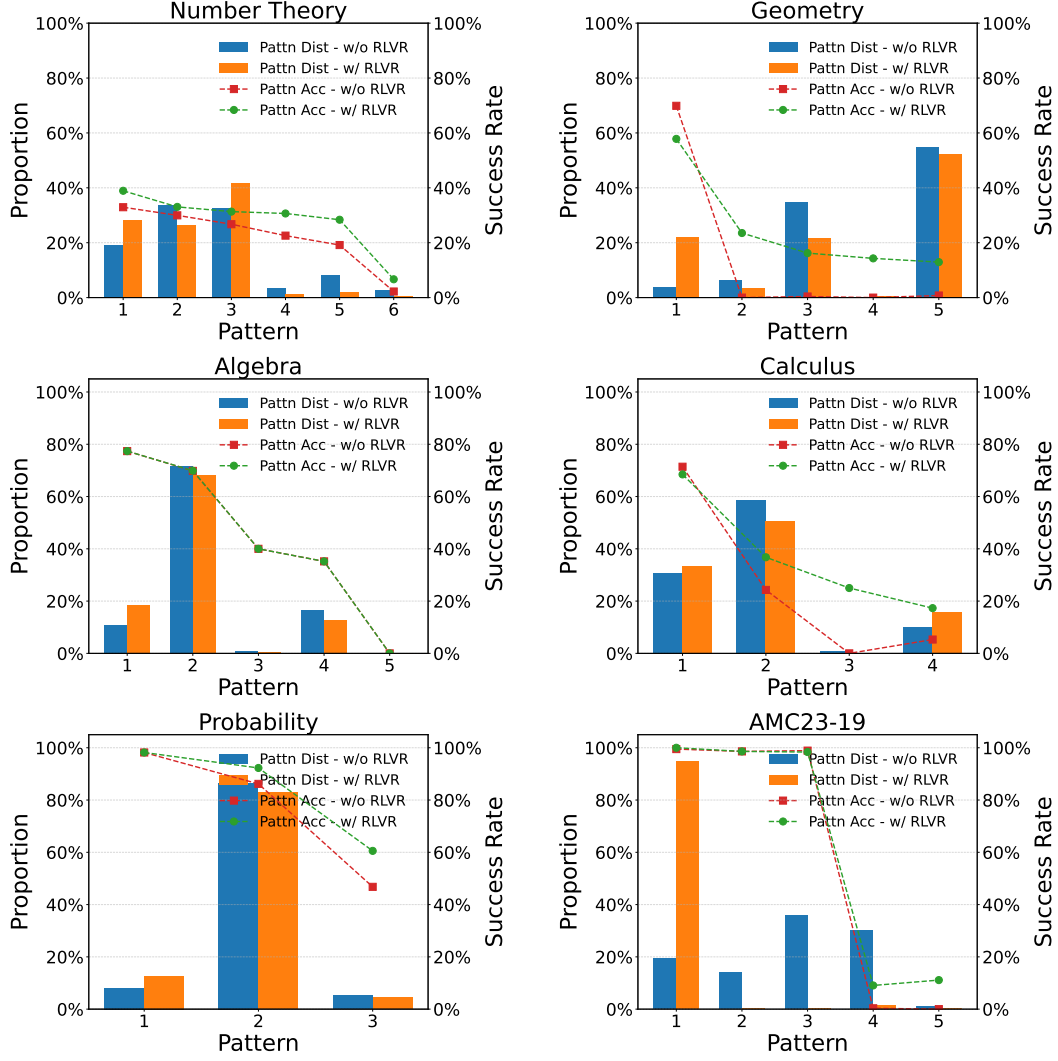
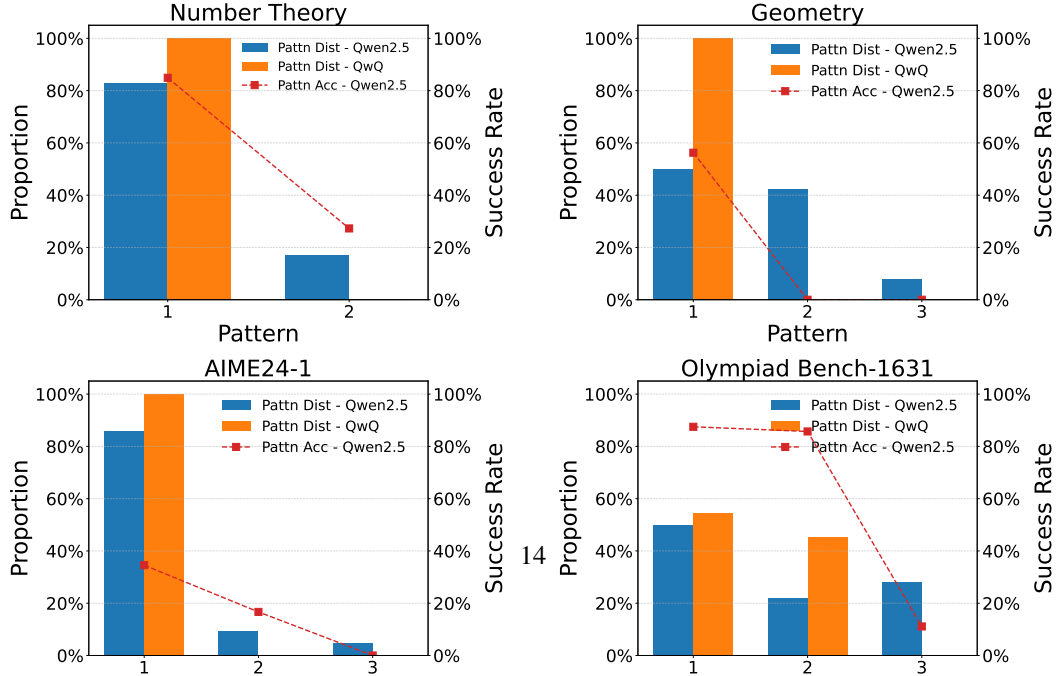


Figure 4: Reasoning patterns and their success rates with and without RLVR enhancement across additional tasks, in close agreement with our experimental insights and theoretical conclusions. The bar (Pattern Dist) gives the proportion of a given pattern among all patterns; the dot (Pattern Acc) gives its accuracy, or success rate.



B Experimental Details

B.1 Experimental Setup

Entire Training Procedures: We first select Qwen2.5-3B as the base model and train it on the MATH dataset, using both RLVR and RLIF. We adopt the verl [31] framework and ensure that all parameter settings are identical to the example on the MATH dataset provided by the verl framework. For RLIF, we only modified the reward calculation method. By setting different random seeds, we obtain three different training curves for each method.

Token-level Analysis: We use Qwen2.5-3B and the above RL-enhanced models for this experiment. For GSM8K and MATH, we randomly sample 400 questions from their test sets respectively. For AIME24, we use its full dataset. We first sample the corresponding answer from the base model for each question. Next, we concatenate this answer to the question itself and input it into the RL-enhanced model as a new prompt. We use the interface in the OpenAI library to calculate the probability of each token in the prompt and its ranking among the probabilities of all possible tokens at the current position. By comparing whether the ranking of each token has changed before and after RL, we can calculate the proportion of tokens whose ranking has changed.

Reasoning-pattern Level Analysis: For this task, we conduct two types of experiments. We first study the dynamics of the patterns distribution during training using previously trained models, namely Qwen2.5-3B and RL-enhanced Qwen2.5-3B. We evaluate every 20 steps. To examine the applicability of our conclusions on different base models and larger models, we test the open-source model Qwen-2.5-7B-SimpleRL-Zoo, QwQ-32B, and their corresponding base models. Therefore, we can only examine the changes in the patterns distribution of model outputs before and after RL, but cannot examine the dynamics of the patterns distribution during the training process.

B.2 Reasoning Patterns Analysis Pipeline

Model Inference: We first randomly sample questions of different task types (such as Geometry, Algebra, etc.) and different difficulty levels (Level 4, Level 5) from datasets including MATH. For each question, our model samples 1024 answers in the patterns distribution dynamic task, and samples answers ranging from 64 to 1024 in the task that only considers the distribution change before and after RL, depending on the task category. These responses will be used for subsequent pattern extraction and classification.

Reasoning Patterns Extraction: We sample a subset of responses of the base model and use GPT-4o [19] API to summarize patterns categories from these samples. Full prompt we used for reasoning patterns extraction is given in Appendix D.1. To ensure the accuracy of patterns extraction, we set the temperature of GPT-4o’s API to 0.

Responses Classification: We again employ GPT-4o to classify all responses according to the identified pattern categories, also setting the temperature to 0. Full prompt we used for responses classification is given in Appendix D.2.

Accuracy Analysis: In order to only consider the impact of the patterns distribution on accuracy, we construct new prompts by concatenating the original prompt with partial responses from the base model, assuming these partial responses sufficiently represent specific reasoning patterns. This ensures that the reasoning process is fixed, which makes it easier for us to determine whether the change in model accuracy depends only on the change in the distribution of the reasoning pattern, rather than the change in the reasoning process itself. Specifically, we remove the sentence containing the final answer from the base model’s response, add the sentence before it to the end of the question, and input it as a new prompt to the RL-enhanced model. The RL-enhanced model will continue to predict the answer. We will examine the accuracy of the predicted answer as the accuracy of the reasoning pattern.

Task	Reasoning Patterns Examples	Common Elements / Key Words
Number Theory	Modular Congruence with Coefficient Simplification	Uses modular arithmetic to express divisibility conditions.
	Inequality-Driven Search for Minimal Solution	Often involves substituting back to compute the total coins.
Geometry	Systematic Inequality Application	Explicitly lists and solves each of the three inequalities.
	Verification of Scalene Condition	Adding an extra layer of validation to verifies the third side length.
Algebra	Iterative Floor Division with Leftover Tracking.	Uses floor division to compute new cans per step.
	Recursive Recycling with Aggregated Leftovers.	Combines leftovers with newly produced cans before recycling.
Calculus	Direct Simplification and Principal Value Matching	Assumes inputs fall within the principal range.
	Interval Analysis with Case Splitting	Splits the whole domain into different intervals.
Counting and Probability	Direct Probability Setup and Quadratic Solution	Explicitly calculates combinations for total and favorable outcomes.
	Early Simplification and Cross-Multiplication	Early cross-multiplication to eliminate denominators.
AMC23-Q19	Prime Factorization and Simplification	Counting Digits in Large Numbers.
	Rewriting (8^5) as (2^{15}) (15^5) as $(3^5 \cdot 5^5)$.	Final step of counting digits: $(3 \text{ (from 243)} + 15 \text{ (zeros)} = 18)$.
AIME24-Q1	Direct Equation Setup and Elimination	Straightforward and relies on algebraic manipulation.
	Alternative Equation Formulation and Solving	Expressing variables in terms of others early on.
OlympiadBench-Q1631	Direct Calculation and Empirical Testing	Relying on direct computation and empirical verification.
	Factorization-Based Reasoning	Using algebraic factorization to argue certain terms y_n .

Table 2: Reasoning Patterns Examples for Varying Tasks.

B.3 Details for Each Task

Below we detail the experimental procedures for each task, including the task descriptions and examples of reasoning patterns (Table 2). Example task prompts are provided in Appendix D.3.

Number Theory: This task presents models with problems involving coin distribution across multiple bags. Initially, bags contain equal numbers of coins. After receiving additional coins and redistributing them equally, the total must exceed a specified value while maintaining equal distribution. Models must determine the minimum initial coin count per bag. We derived this task from a level 5 MATH dataset problem (id: test/number_theory/1055) [16], creating 32 variants by adjusting parameters like bag count and coin totals as our evaluation dataset. We employ one-shot prompting with a simple, unrelated example to guide answer formatting using "boxed" notation without influencing reasoning approaches. For evaluation, we first choose 20 questions from our evaluation dataset with 4 responses each for reasoning pattern extraction. For each question we sample 64 responses. We then categorize all responses by the extracted reasoning patterns, allowing us to compare changes in the distribution of models' reasoning patterns before and after RL.

Geometry: For the geometry task, we will give the model the lengths of two sides of a scalene triangle and ask how many different integer centimeters the length of the third side can be. This task comes from a level 4 geometry problem in MATH (id: test/geometry/1046). Our evaluation follows the same pipeline as in Number Theory: we construct a synthetic dataset with 32 questions, sample 20 questions and select 4 responses each from the base model (Qwen2.5-7b-Instruct) for reasoning pattern extraction, and then we compare the responses for model with and without RLVR enhancement with all questions in our evaluation dataset with 64 responses each for pattern analysis.

Algebra: The algebra task we use is a can recycling problem. We first have a certain number of old cans. It is pre-defined that n old cans can be recycled into a new can. The question is how many cans can be produced in the end. The difficulty of the problem is that in each step of the iterative calculation, there may be extra cans that cannot be divided evenly. These cans may eventually be combined together for further recycling. The template for this task comes from a level 4 algebra problem in MATH (id: test/algebra/2768). For all the following supplementary tasks starting from this task, we adopt the same settings as the previous tasks, including the data set size, number of samples, etc.

Calculus: Our calculus task is simple and straightforward. We will present a trigonometric equation in a single variable, the domain of that variable, and ask the model to determine the number of solutions to the equation. Although this problem is simple, it can demonstrate the model’s basic ability in calculus problems. Its prototype is a level 4 pre-calculus problem in MATH (id: test/precalculus/1140).

Counting and Probability: We also study the performance of the model on the counting and probability task. There are white balls and black balls. We will randomly sample two balls from these balls and give the probability that one of the two balls drawn is black and the other is white. We also provide the number of balls of a certain color and hope that the model can calculate the minimum number of balls of another color. This is a probability theory task, which comes from a level 4 counting and probability problem in MATH (id: test/counting_and_probability/79).

AMC23 Question: Here we choose question 19 in AMC23 [1], for reasoning pattern extraction, we randomly sample 48 responses, and then sample 1024 response for reasoning pattern analysis.

B.4 Details for Experiments on QwQ-32B and Qwen-2.5-32B

We conduct experiments on four tasks. Due to the limitations of using the API for testing, we randomly select one question per task as input, sample 64 responses from each of the two models, and directly perform reasoning patterns extraction and classification from these answers. Specifically, for the two tasks of the MATH dataset, we choose question 1055 from the number theory task and question 1046 from the geometry task. For the remaining two tasks, we question problem 1 from AIME24 and question 1631 from Olympiad Bench. Examples of patterns are shown in Table 2.

C Deferred Proofs

In this appendix, we provide proofs for our main theoretical results: Proposition 5.2, Theorems 5.3, Theorems 5.4 and Theorems 5.6.

C.1 Proof of Proposition 5.2

Proof of Proposition 5.2. In this proof, we utilize the proof techniques in [26], recall that the optimization objective of RL is

$$\phi_{\text{RL}}(\theta) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}, \mathbf{y} \sim \pi_{\theta}(\mathbf{y}|\mathbf{x})} [r_{\phi}(\mathbf{x}, \mathbf{y})] - \beta \mathbb{D}_{\text{KL}} [\pi_{\theta}(\mathbf{y} | \mathbf{x}) \| \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})], \quad (\text{C.1})$$

Under our framework, as state in section 5.1, we consider a policy conduct reasoning by first sample $\mathbf{r}_i \in \mathcal{R}$ based on $\pi_{\theta}(\mathbf{r}_i|\mathbf{q})$ and then provide the final answer $\mathbf{a} \in \mathcal{A}$ by $\pi_{\theta}(\mathbf{a}|\mathbf{r}_i)$, the reward $\phi_{\text{RL}}(\theta)$ can be written as

$$\phi_{\text{RL}}(\theta) = \mathbb{E}_{\mathbf{y} \sim \pi_{\theta}(\mathbf{r}|\mathbf{q})} \left[\sum_{\mathbf{a} \in \mathcal{A}} \pi_{\text{ref}}(\mathbf{a} | \mathbf{r}) r_{\phi}([\mathbf{r}, \mathbf{a}], \mathbf{q}) \right] - \beta \mathbb{D}_{\text{KL}} [\pi_{\theta}(\mathbf{y} | \mathbf{x}) \| \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})], \quad (\text{C.2})$$

644 Let $R(\mathbf{r}) = \sum_{\mathbf{a} \in \mathcal{A}} \pi_{\text{ref}}(\mathbf{a} \mid \mathbf{r}) r_\phi([\mathbf{r}, \mathbf{a}], q)$, then

$$\begin{aligned}
\phi_{\text{RL}}(\boldsymbol{\theta}) &= \mathbb{E}_{\mathbf{r} \sim \pi_{\boldsymbol{\theta}}(\cdot \mid \mathbf{q}), \mathbf{a} \sim \pi_{\boldsymbol{\theta}}(\cdot \mid \mathbf{r})} \left[R(\mathbf{r}) - \beta \ln \left(\frac{\pi_{\boldsymbol{\theta}}(\mathbf{r} \mid \mathbf{q}) \cdot \pi_{\text{ref}}(\mathbf{a} \mid \mathbf{r})}{\pi_{\text{ref}}(\mathbf{r} \mid \mathbf{q}) \cdot \pi_{\text{ref}}(\mathbf{a} \mid \mathbf{r})} \right) \right] \\
&= \mathbb{E}_{\mathbf{r} \sim \pi_{\boldsymbol{\theta}}(\cdot \mid \mathbf{q})} \left[R(\mathbf{r}) - \beta \ln \left(\frac{\pi_{\boldsymbol{\theta}}(\mathbf{r} \mid \mathbf{q})}{\pi_{\text{ref}}(\mathbf{r} \mid \mathbf{q})} \right) \right] \\
&= -\beta \mathbb{E}_{\mathbf{r} \sim \pi_{\boldsymbol{\theta}}(\cdot \mid \mathbf{q})} \left[\ln \left(\frac{\pi_{\boldsymbol{\theta}}(\mathbf{r} \mid \mathbf{q})}{\frac{1}{Z} \exp \left(\frac{1}{\beta} R(\mathbf{r}) \right) \pi_{\text{ref}}(\mathbf{r} \mid \mathbf{q})} \right) - \ln Z \right] \\
&= -\beta \mathbb{D}_{\text{KL}} [\pi^*(\mathbf{r} \mid \mathbf{q}) \parallel \pi_{\text{ref}}(\mathbf{r} \mid \mathbf{q})] + \beta \ln Z.
\end{aligned}$$

645 Where the third equation is by Assumption 5.1 and $Z = \sum_{\mathbf{r} \in \mathcal{R}} \exp \left(\frac{1}{\beta} p^*(\mathbf{r}) \right) \pi_{\boldsymbol{\theta}_{\text{ref}}}(\mathbf{r} \mid \mathbf{q})$ is the
646 partition constant that ensures

$$\pi^*(\mathbf{r} \mid \mathbf{q}) = \frac{1}{Z} \exp \left(\frac{1}{\beta} R(\mathbf{r}) \right) \pi_{\boldsymbol{\theta}}(\mathbf{r} \mid \mathbf{q}),$$

647 is a valid probability distribution such that $\sum_{\mathbf{r} \in \mathcal{R}} \pi^*(\mathbf{r} \mid \mathbf{q}) = 1$. Since Z is not a function of $\mathbf{r}$.

648 Therefore, maximizing the objective in Equation C.2 is equivalent to:

$$\begin{aligned}
&\max_{\pi} \mathbb{E}_{\mathbf{x} \sim \mathcal{D}, \mathbf{y} \sim \pi(\mathbf{y} \mid \mathbf{x})} [r_\phi(\mathbf{x}, \mathbf{y})] - \beta \mathbb{D}_{\text{KL}} [\pi(\mathbf{y} \mid \mathbf{x}) \parallel \pi_{\text{ref}}(\mathbf{y} \mid \mathbf{x})] \\
&= \min_{\pi} \beta \mathbb{D}_{\text{KL}} [\pi^*(\mathbf{r} \mid \mathbf{q}) \parallel \pi_{\text{ref}}(\mathbf{r} \mid \mathbf{q})] - \beta \ln Z.
\end{aligned}$$

649 By the properties of KL-divergence, we know that the optimal policy for the KL-constrained reward
650 maximization objective satisfies:

$$\pi_{\text{opt}}(\mathbf{r} \mid \mathbf{q}) = \frac{1}{Z} \exp \left(\frac{1}{\beta} R(\mathbf{r}) \right) \pi_{\boldsymbol{\theta}_{\text{ref}}}(\mathbf{r} \mid \mathbf{q}) \text{ for all } \mathbf{r} \in \mathcal{R}.$$

651 specifically, for RLVR, $R_{\text{RLVR}}(\mathbf{r}) = p^*$, for RLIF, we have

$$\begin{aligned}
R'(\mathbf{r}) &= - \left(\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_i \in \mathcal{R}} \log (|\mathcal{R}| \cdot \pi_{\boldsymbol{\theta}}(\mathbf{r}_i \mid \mathbf{q})) + \frac{1}{|\mathcal{A}|} \sum_{\mathbf{a}_j \in \mathcal{A}} \log (|\mathcal{A}| \cdot \pi_{\boldsymbol{\theta}}(\mathbf{a}_j \mid \mathbf{q}, \mathbf{r})) \right) \\
&= - \frac{1}{|\mathcal{A}|} \sum_{\mathbf{a}_j \in \mathcal{A}} \log (\pi_{\boldsymbol{\theta}}(\mathbf{a}_j \mid \mathbf{q}, \mathbf{r})) - \underbrace{\left(\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_i \in \mathcal{R}} \log (|\mathcal{R}| \cdot \pi_{\boldsymbol{\theta}}(\mathbf{r}_i \mid \mathbf{q})) + \log |\mathcal{A}| \right)}_c,
\end{aligned}$$

652 where the second term is the same for all $\mathbf{r} \in |\mathcal{R}|$, for RLIF, the optimal policy can be written as

$$\begin{aligned}
\pi_{\text{opt}}(\mathbf{r} \mid \mathbf{q}) &= \frac{1}{Z} \exp \left(\frac{1}{\beta} R'(\mathbf{r}) \right) \pi_{\boldsymbol{\theta}_{\text{ref}}}(\mathbf{r} \mid \mathbf{q}) \\
&= \frac{\exp \left(\frac{1}{\beta} R'(\mathbf{r}) \right) \pi_{\boldsymbol{\theta}_{\text{ref}}}(\mathbf{r} \mid \mathbf{q})}{\sum_{\mathbf{r}_i \in \mathcal{R}} \exp \left(\frac{1}{\beta} (R'(\mathbf{r}_i)) \right) \pi_{\boldsymbol{\theta}_{\text{ref}}}(\mathbf{r}_i \mid \mathbf{q})} \\
&= \frac{\exp \left(\frac{1}{\beta} (R'(\mathbf{r}) - c) \right) \pi_{\boldsymbol{\theta}_{\text{ref}}}(\mathbf{r} \mid \mathbf{q})}{\sum_{\mathbf{r}_i \in \mathcal{R}} \exp \left(\frac{1}{\beta} (R'(\mathbf{r}_i) - c) \right) \pi_{\boldsymbol{\theta}_{\text{ref}}}(\mathbf{r}_i \mid \mathbf{q})}
\end{aligned}$$

653 So we can write $R_{\text{RLIF}}(\mathbf{r}) = R'(\mathbf{r}) - c = -\frac{1}{|\mathcal{A}|} \sum_{\mathbf{a}_j \in \mathcal{A}} \log (\pi_{\boldsymbol{\theta}}(\mathbf{a}_j \mid \mathbf{q}, \mathbf{r}))$. This concludes the proof
654 of the theorem. $\square$

655 C.2 Dynamics of Tabular Policy

656 Consider the RL objective (Eq 3.1), using a tabular policy (Eq 5.3) with Assumption 5.1 holds, the
 657 gradient dynamics of θ can be computed as

$$\begin{aligned}\frac{d}{dt}\theta(t) &= \nabla\phi_{\text{RL}}(\theta(t)) \\ &= \nabla\mathbb{E}_{\mathbf{r} \sim \pi_{\theta}(\cdot|\mathbf{q})} [r_{\phi}^{KL}(\mathbf{r}|\mathbf{q})] \\ &= \sum_{\mathbf{r} \in \mathcal{R}} [r_{\phi}^{KL}(\mathbf{r}|\mathbf{q}) \nabla\pi_{\theta(t)}(\mathbf{r}|\mathbf{q})],\end{aligned}$$

658 Where $\pi_{\theta(t)}(\mathbf{r}|\mathbf{q}) = \text{softmax}(\theta_{:,q})_{\mathbf{r}}$, $\theta_{:,q} \in \mathbb{R}^{|\mathcal{V}|}$ is the column of θ corresponding to $\mathbf{q}$, and
 659 $r_{\phi}(\mathbf{r}|\mathbf{q})^{KL}$ refers to the reward with KL divergence, so

$$\begin{aligned}\frac{\partial\theta_{\mathbf{r}_i,\mathbf{q}}(t)}{\partial t} &= \sum_{\mathbf{r}_j \in \mathcal{R}} \left[r_{\phi}^{KL}(\mathbf{r}_j|\mathbf{q}) (\nabla\text{softmax}(\theta_{:,q})_{\mathbf{r}_j})_{\mathbf{r}_i} \right] \\ &= \sum_{\mathbf{r}_j \in \mathcal{R}} \left[r_{\phi}^{KL}(\mathbf{r}_j|\mathbf{q}) (-\pi_{\theta(t)}(\mathbf{r}_j|\mathbf{q}) \cdot \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})) \right] \\ &\quad + r_{\phi}^{KL}(\mathbf{r}_i|\mathbf{q}) \cdot \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})\end{aligned}$$

660 where the overall accuracy is defined as $\text{ACC}_{\theta(t)} = \sum_{\mathbf{r} \in \mathcal{R}} \pi_{\theta(t)}(\mathbf{r}|\mathbf{q}) p^*(\mathbf{r})$. For $\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})$, we
 661 have

$$\frac{d}{dt}\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) = \sum_{\mathbf{r}_j \in \mathcal{R}} \frac{\partial\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})}{\partial\theta_{\mathbf{r}_j,\mathbf{q}}(t)} \cdot \frac{\partial\theta_{\mathbf{r}_j,\mathbf{q}}(t)}{\partial t}$$

662 C.3 Proof of Theorem 5.3

663 To proof Theorem 5.3, here we first prove that $\text{ACC}_{\theta(t)} \geq \text{ACC}_{\theta(0)}$ for $t \geq 0$ (Eq C.4), then we
 664 establish a lower bound for $\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})$ (Eq C.5), finally we derive the final bound for t stated in
 665 Theorem 5.3.

666 *Proof of Theorem 5.3.* Setting $\beta = 0$, Eq C.3 becomes:

$$\frac{\partial\theta_{\mathbf{r}_i,\mathbf{q}}(t)}{\partial t} = \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \cdot (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}). \quad (\text{C.3})$$

667 For $\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})$, we derive:

$$\begin{aligned}\frac{d}{dt}\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) &= \sum_{\mathbf{r}_j \in \mathcal{R}} \frac{\partial\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})}{\partial\theta_{\mathbf{r}_j,\mathbf{q}}(t)} \cdot \frac{\partial\theta_{\mathbf{r}_j,\mathbf{q}}(t)}{\partial t} \\ &= \sum_{\mathbf{r}_j \in \mathcal{R}} (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \\ &\quad + (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}_i|\mathbf{q}),\end{aligned}$$

668 and for the accuracy $\text{ACC}_{\theta(t)}$:

$$\begin{aligned}
\frac{d}{dt} \text{ACC}_{\theta(t)} &= \sum_{\mathbf{r}_i \in \mathcal{R}} p^*(\mathbf{r}_i) \cdot \frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}_i | \mathbf{q}) \\
&= \sum_{\mathbf{r}_i, \mathbf{r}_j \in \mathcal{R}} p^*(\mathbf{r}_i) \cdot (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j | \mathbf{q}) \pi_{\theta(t)}(\mathbf{r}_i | \mathbf{q}) \\
&\quad + \sum_{\mathbf{r}_i \in \mathcal{R}} (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}_i | \mathbf{q}) \\
&= \sum_{\mathbf{r}_j \in \mathcal{R}} \text{ACC}_{\theta(t)} \cdot (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j | \mathbf{q}) \\
&\quad + \sum_{\mathbf{r}_i \in \mathcal{R}} (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}_i | \mathbf{q}) p^*(\mathbf{r}_i) \\
&= \sum_{\mathbf{r}_i \in \mathcal{R}} \pi_{\theta(t)}^2(\mathbf{r}_i | \mathbf{q}) (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)})^2 \geq 0. \tag{C.4}
\end{aligned}$$

669 As $\frac{d}{dt} \text{ACC}_{\theta(t)} \geq 0$, so $\text{ACC}_{\theta(t)} > \text{ACC}_{\theta(0)}$ holds for $t \geq 0$, for the optimal reasoning pattern
670 $\mathbf{r}^*$, let $\mathbf{r}'$ the suboptimal reasoning pattern with the second highest success rate, defined as $\mathbf{r}' =$
671 $\arg \max_{\mathbf{r}, \mathbf{r} \neq \mathbf{r}^*} p^*(\mathbf{r})$, we have:

$$\begin{aligned}
\frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}) &= \sum_{\mathbf{r}_j \in \mathcal{R}} (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j | \mathbf{q}) \pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}) \\
&\quad + (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}^* | \mathbf{q}) \\
&\geq (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}^*)) \pi_{\theta(t)}^2(\mathbf{r}^* | \mathbf{q}) \pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}) \\
&\quad + (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}^* | \mathbf{q}) \\
&= (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}^* | \mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q})) \\
&\geq (p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) \pi_{\theta(t)}^2(\mathbf{r}^* | \mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}))^2 \geq 0. \tag{C.5}
\end{aligned}$$

672 As $\frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}) \geq 0$, $\pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}) \geq \pi_{\theta(0)}(\mathbf{r}^* | \mathbf{q})$ holds for $t \geq 0$, define $C =$
673 $(p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) \pi_{\theta(0)}^2(\mathbf{r}^* | \mathbf{q})$:

$$\frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}) \geq C (1 - \pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}))^2.$$

674 This differential inequality yields:

$$\pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}) > 1 - \frac{1}{Ct + \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}^* | \mathbf{q})}}.$$

675 For any $\epsilon > 0$, there exists $T_1 = \frac{1}{C} \left(\frac{1}{\epsilon} - \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}^* | \mathbf{q})} \right) = \mathcal{O}(\frac{1}{\epsilon})$, such that

$$\pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}) > 1 - \epsilon.$$

676 This concludes the proof of the theorem. □

677 C.4 Proof of Theorem 5.4

678 To proof Theorem 5.4, we first establish an upper bound for $\pi_{\theta(t)}(\mathbf{r}' | \mathbf{q})$ (Eq C.6), then we analyze
679 the ratio $\rho_{\theta(t)}(\mathbf{r}_i) = \frac{\pi_{\theta(t)}(\mathbf{r}_i | \mathbf{q})}{\pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q})}$ (Eq C.7) and prove that for $t > T_0$, $\text{ACC}_{\theta(t)} \geq p^*(\mathbf{r}')$.

680 *Proof of Theorem 5.4.* Let $\mathbf{r}^*$ and $\mathbf{r}'$ be the optimal and second optimal reasoning patterns. For
 681 $\pi_{\theta(t)}(\mathbf{r}'|\mathbf{q})$:

$$\begin{aligned}
 \frac{d}{dt}\pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) &= \sum_{\mathbf{r}_j \in \mathcal{R}} (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) \\
 &\quad + (p^*(\mathbf{r}') - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}'|\mathbf{q}) \\
 &= \sum_{\mathbf{r}_j \in \mathcal{R}/\{\mathbf{r}^*\}} (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) \\
 &\quad + (p^*(\mathbf{r}') - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}'|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})) \\
 &\leq \sum_{\mathbf{r}_j \in \mathcal{R}/\{\mathbf{r}^*\}} \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) \\
 &\quad + (p^*(\mathbf{r}') - p^*(\mathbf{r}') \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})) \pi_{\theta(t)}^2(\mathbf{r}'|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})) \\
 &\leq \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}))^2 + p^*(\mathbf{r}') \pi_{\theta(t)}^2(\mathbf{r}'|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}))^2 \\
 &\leq 2 (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}))^2.
 \end{aligned}$$

682 This yields:

$$\pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) \leq 1 - \frac{1}{2t + \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})}}. \quad (\text{C.6})$$

683 Next, define $\rho_{\theta(t)}(\mathbf{r}_i) = \frac{\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})}{\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})}$, then for all $\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}^*, \mathbf{r}'\}$, we have

$$\begin{aligned}
 \frac{d}{dt}\rho_{\theta(t)}(\mathbf{r}_i) &= \frac{\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \frac{d}{dt}\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) - \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \frac{d}{dt}\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})}{\pi_{\theta(t)}^2(\mathbf{r}^*|\mathbf{q})} \\
 &= \frac{\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})}{\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})} ((p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) - (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})) \\
 &\leq -\rho_{\theta(t)}(\mathbf{r}_i) (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \leq 0
 \end{aligned}$$

684 The last inequality is based on the condition for case 2 and $\text{ACC}_{\theta(t)} > \text{ACC}_{\theta(0)}$, so $p^*(\mathbf{r}_i) <$
 685 $\text{ACC}_{\theta(t)}$ holds for all $\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}^*\}$, $t \geq 0$. Then as $\frac{d}{dt}\rho_{\theta(t)}(\mathbf{r}_i) \leq 0$, we have:

$$\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) = \frac{1 - \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q})}{\sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(t)}(\mathbf{r}_i)} \geq \frac{1 - \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q})}{\sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(0)}(\mathbf{r}_i)}$$

686 While $\text{ACC}_{\theta(t)} < p^*(\mathbf{r}')$, define $\gamma_{\pi_{\text{ref}}} := \sum_{\mathbf{r} \in \mathcal{R}/\{\mathbf{r}'\}} \frac{\pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q})}{\pi_{\theta_{\text{ref}}}(\mathbf{r}^*|\mathbf{q})}$ we have:

$$\begin{aligned}
 \frac{d}{dt}\rho_{\theta(t)}(\mathbf{r}_i) &\leq -\rho_{\theta(t)}(\mathbf{r}_i) (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \\
 &\leq -\rho_{\theta(t)}(\mathbf{r}_i) (p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) \frac{1 - \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q})}{\gamma_{\pi_{\text{ref}}}} \\
 &\leq -\rho_{\theta(t)}(\mathbf{r}_i) \frac{p^*(\mathbf{r}^*) - p^*(\mathbf{r}')}{\gamma_{\pi_{\text{ref}}}} \frac{1}{2t + \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})}}
 \end{aligned}$$

687 Then we have:

$$\frac{d}{dt}\rho_{\theta(t)}(\mathbf{r}_i) \leq -\rho_{\theta(t)}(\mathbf{r}_i) \frac{p^*(\mathbf{r}^*) - p^*(\mathbf{r}')}{\gamma_{\pi_{\text{ref}}}} \frac{1}{2t + \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})}}, \quad (\text{C.7})$$

688 this differential inequality leads to:

$$\rho_{\theta(t)}(\mathbf{r}_i) \leq \frac{(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q}))^{-\Delta/(2\gamma_{\pi_{\text{ref}}})}}{(1/(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})) + 2t)^{\Delta/(2\gamma_{\pi_{\text{ref}}})}} \rho_{\theta(0)}(\mathbf{r}_i).$$

689 Where $\Delta = p^*(\mathbf{r}^*) - p^*(\mathbf{r}')$. Recall that:

$$\begin{aligned} \text{ACC}_{\theta(t)} - p^*(\mathbf{r}') &= \sum_{\mathbf{r} \in \mathcal{R}} \pi_{\theta(t)}(\mathbf{r}|\mathbf{q}) p^*(\mathbf{r}) - p^*(\mathbf{r}') \\ &= p^*(\mathbf{r}^*) \left[(p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) + \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(t)}(\mathbf{r}_i) (p^*(\mathbf{r}_i) - p^*(\mathbf{r}')) \right] \\ &\geq p^*(\mathbf{r}^*) \left[(p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) - \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(t)}(\mathbf{r}_i) p^*(\mathbf{r}') \right]. \end{aligned}$$

690 Let $C_2 = 1/\Delta, C_1 = p^*(\mathbf{r}')/\Delta$, define:

$$T_0 = \frac{1}{2 - 2\pi_{\theta_{\text{ref}}}(\mathbf{r}'|\mathbf{q})} \left((C_1 \cdot \gamma_{\pi_{\text{ref}}})^{2C_2 \cdot \gamma_{\pi_{\text{ref}}}} - 1 \right),$$

691 then:

$$\begin{aligned} &\sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(T_0)}(\mathbf{r}_i) p^*(\mathbf{r}') \\ &\leq \frac{(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q}))^{-\Delta/(2\gamma_{\pi_{\text{ref}}})}}{(1/(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})) + 2T_0)^{\Delta/(2\gamma_{\pi_{\text{ref}}})}} \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(0)}(\mathbf{r}_i) p^*(\mathbf{r}') \\ &\leq \frac{(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q}))^{-\Delta/(2\gamma_{\pi_{\text{ref}}})}}{\left(1/(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})) \cdot \left(\frac{\gamma_{\pi_{\text{ref}}} p^*(\mathbf{r}')}{\Delta} \right)^{2\gamma_{\pi_{\text{ref}}}/\Delta} \right)^{\Delta/(2\gamma_{\pi_{\text{ref}}})}} p^*(\mathbf{r}') \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(0)}(\mathbf{r}_i) \\ &= \frac{1}{\left(\left(\frac{\gamma_{\pi_{\text{ref}}} p^*(\mathbf{r}')}{\Delta} \right)^{2\gamma_{\pi_{\text{ref}}}/\Delta} \right)^{\Delta/(2\gamma_{\pi_{\text{ref}}})}} p^*(\mathbf{r}') \gamma_{\pi_{\text{ref}}} \\ &= \Delta = p^*(\mathbf{r}^*) - p^*(\mathbf{r}'). \end{aligned}$$

692 Finally, for $t > T_0$ we have:

$$\begin{aligned} \text{ACC}_{\theta(t)} - p^*(\mathbf{r}') &\geq p^*(\mathbf{r}^*) \left[(p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) - \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(t)}(\mathbf{r}_i) p^*(\mathbf{r}') \right] \\ &\geq p^*(\mathbf{r}^*) [(p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) - (p^*(\mathbf{r}^*) - p^*(\mathbf{r}'))] = 0. \end{aligned}$$

693 This concludes the proof of the theorem. $\square$

694 C.5 Proof of Theorem 5.6

695 *Proof of Theorem 5.6.* First, as we consider the case $|\mathcal{A}| = 2$ and $\pi_{\theta_{\text{base}}}(\mathbf{r}|\mathbf{q}) = \frac{1}{|\mathcal{R}|}$, Assumption
696 5.5 holds means $\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r} \in \mathcal{R}} (p^*(\mathbf{r})) > 0.5$. The derivative of overall accuracy can be written as

$$\begin{aligned} \frac{d}{dt} \text{ACC}_{\theta(t)} &= \sum_{\mathbf{r}_i \in \mathcal{R}} p^*(\mathbf{r}_i) \cdot \frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \\ &= \sum_{\mathbf{r}_i \in \mathcal{R}} \pi_{\theta(t)}^2(\mathbf{r}_i|\mathbf{q}) (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \left(- \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(t)}(\mathbf{a}|\mathbf{r}_i, \mathbf{q})) - \text{Con}_{\theta(t)} \right), \end{aligned}$$

697 where $\text{Con}_{\theta(t)} = -\sum_{\mathbf{r}_i \in \mathcal{R}} \pi_{\theta(t)}(\mathbf{r}_i | \mathbf{q}) \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(t)}(\mathbf{a} | \mathbf{r}_i, \mathbf{q}))$. Here we assume $\mathcal{A} = \{\mathbf{a}^*, \mathbf{a}'\}$
 698 where $\mathbf{a}'$ is the incorrect answer, and $\pi_{\theta(0)}(\mathbf{r}_i | \mathbf{q}) = \frac{1}{|\mathcal{R}|}$ for all $\mathbf{r}_i \in \mathcal{R}$. Then:

$$\begin{aligned} \left. \frac{d}{dt} \text{ACC}_{\theta(t)} \right|_{t=0} &= \sum_{\mathbf{r}_i \in \mathcal{R}} \frac{1}{|\mathcal{R}|^2} \left(p^*(\mathbf{r}_i) - \frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_j \in \mathcal{R}} p^*(\mathbf{r}_j) \right) \cdot \\ &\quad \left(-\sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(0)}(\mathbf{a} | \mathbf{r}_i, \mathbf{q})) + \frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_j \in \mathcal{R}} \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(0)}(\mathbf{a} | \mathbf{r}_j, \mathbf{q})) \right) \\ &= \frac{1}{|\mathcal{R}|^2} \sum_{\mathbf{r}_i \in \mathcal{R}} \left(p^*(\mathbf{r}_i) - \frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_j \in \mathcal{R}} p^*(\mathbf{r}_j) \right) \cdot \left(-\sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(0)}(\mathbf{a} | \mathbf{r}_i, \mathbf{q})) \right). \end{aligned}$$

699 As we consider the case $|\mathcal{R}| \rightarrow +\infty$ and the success rate $p^*(\mathbf{r})$ are i.i.d. samples from $U[0, 1]$.
 700 By the Central limit theorem, for large $|\mathcal{R}|$, the sample mean $\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r} \in \mathcal{R}} p^*(\mathbf{r})$ is concentrated
 701 around the its expectation 0.5, the deviation is of order is $\mathcal{O}(1/\sqrt{|\mathcal{R}|})$. We define this deviation as
 702 $\delta = \mathcal{O}(1/|\mathcal{R}|^{1/2}) > 0$ such that $\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r} \in \mathcal{R}} p^*(\mathbf{r}) = 0.5 + \delta$.

703 According to the Gibbs conditioning principle ([12] Corollary 7.3.5 and Theorem 7.3.8), as $|\mathcal{R}| \rightarrow \infty$,
 704 the empirical distribution of the $p^*(\mathbf{r})$ conditioned on their mean $\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r} \in \mathcal{R}} p^*(\mathbf{r}) = 0.5 + \delta$
 705 converges to the density: $f(x) = 1 + 12\delta(x - 0.5), x \in [0, 1]$. As $|\mathcal{R}| \rightarrow \infty$, we can approximate
 706 the summation over $\mathbf{r}$ with an integral over this limiting distribution $f(x)$. The derivative becomes:

$$\left. \frac{d}{dt} \text{ACC}_{\theta(t)} \right|_{t=0} = \int_0^1 - (1 + 12\delta(x - 0.5)) \cdot (x - 0.5 - \delta) \cdot \log(x(1 - x)) dx = \frac{2}{3} \delta > 0$$

707 Since $\delta > 0$, the derivative is positive. As $|\mathcal{R}| \rightarrow \infty$, this holds with probability 1, proving the
 708 first statement.

To prove the second statement, we first analyze the RLIF reward function. For an action space of size
 $|\mathcal{A}| = 2$, the reward for a path $\mathbf{r}$ is proportional to the negative log-likelihood of the policy:

$$R_{\text{RLIF}}(\mathbf{r}) - \frac{1}{|\mathcal{A}|} \sum_{\mathbf{a}_j \in \mathcal{A}} \log(\pi_{\theta}(\mathbf{a}_j | \mathbf{q}, \mathbf{r})) = -\frac{1}{2} \log(p^*(\mathbf{r})(1 - p^*(\mathbf{r})))$$

709 The function $g(p) = p(1 - p)$ is maximized at $p = 0.5$. Therefore, maximizing the reward $R_{\text{RLIF}}(\mathbf{r})$
 710 is equivalent to choosing the path whose success rate $p^*(\mathbf{r})$ is furthest from 0.5.

Let $p_{\min}^* = \min_{\mathbf{r} \in \mathcal{R}} p^*(\mathbf{r})$ and $p_{\max}^* = \max_{\mathbf{r} \in \mathcal{R}} p^*(\mathbf{r})$. The path with the highest reward will be the
 one with success rate $p_{\min}^*$ if it is further from 0.5 than $p_{\max}^*$ is. This condition is expressed as:

$$0.5 - p_{\min}^* > p_{\max}^* - 0.5 \iff p_{\min}^* + p_{\max}^* < 1$$

711 Thus, the event $\arg \max_{\mathbf{r}} R_{\text{RLIF}}(\mathbf{r}) = \arg \min_{\mathbf{r}} p^*(\mathbf{r})$ is equivalent to the event $p_{\min}^* + p_{\max}^* < 1$.

712 We now calculate the probability of this event. The success rates $p^*(\mathbf{r})$ are drawn from the limiting
 713 distribution $f(x) = 1 + 12\delta(x - 0.5)$. From the theory of order statistics, the scaled minimum and
 714 maximum of $n = |\mathcal{R}|$ samples converge in distribution to independent exponential random variables.
 715 Let $Y_n = n \cdot p_{\min}^*$ and $Z_n = n \cdot (1 - p_{\max}^*)$. As $n \rightarrow \infty$:

- 716 • Y_n converges to $Y \sim \text{Exp}(\lambda_Y)$, where the rate is the density at the lower bound: $\lambda_Y = f(0) =$
 717 $1 - 6\delta$.
- 718 • Z_n converges to $Z \sim \text{Exp}(\lambda_Z)$, where the rate is the density at the upper bound: $\lambda_Z = f(1) =$
 719 $1 + 6\delta$.

The condition $p_{\min}^* + p_{\max}^* < 1$ can be rewritten in terms of these asymptotic variables:

$$\frac{Y_n}{n} + \left(1 - \frac{Z_n}{n}\right) < 1 \implies Y_n < Z_n$$

The probability of this event is $P(Y < Z)$. For two independent exponential variables, this is given by:

$$P(Y < Z) = \frac{\lambda_Y}{\lambda_Y + \lambda_Z} = \frac{1 - 6\delta}{(1 - 6\delta) + (1 + 6\delta)} = \frac{1 - 6\delta}{2} = \frac{1}{2} - 3\delta$$

Since $\delta = O(1/|\mathcal{R}|^{1/2})$, in the limit as $|\mathcal{R}| \rightarrow \infty$, $\delta \rightarrow 0$. Therefore, the limiting probability is:

$$\lim_{|\mathcal{R}| \rightarrow \infty} P(\arg \max_{\mathbf{r}} R_{\text{RLIF}}(\mathbf{r}) = \arg \min_{\mathbf{r}} p^*(\mathbf{r})) = \lim_{\delta \rightarrow 0} \left(\frac{1}{2} - 3\delta \right) = \frac{1}{2}$$

720 This concludes the proof of the second statement.

721

□

722 D Full Prompts

723 D.1 Full Prompt for Reasoning Patterns Extraction

Full Prompt for Reasoning Patterns Extraction

You will analyze multiple AI reasoning processes, showing how different models solve problems.

Analysis Steps:

For each reasoning process, identify:

Key words and recurring phrases

Logical structure of the argument

Problem-solving techniques used

Step-by-step progression

Group similar reasoning processes into exactly 5 patterns based on shared:

Vocabulary patterns (common terms and phrases)

Logical frameworks (how arguments are structured)

Solution approaches (methods used to reach conclusions)

IMPORTANT! You need to categorize general thinking patterns for problem solving, rather than categorizing types of problems themselves. In other words, each type of thinking pattern you give can be applied to any problem, rather than being task-specific. What you need to classify is the way of thinking, not the way of solving problems.

For example, you can classify them into:

1. derivation based on existing knowledge
2. derivation combined with verification
3. independent calculation, suitable for learners starting from the basics,
4. detailed manual derivation, suitable for learners who need to deeply understand each step
5. etc

Or you can classify them into:

1. knowledge call type
2. derivation and verification type
3. etc.

Output Format:

Pattern 1:

Name and description

Defining characteristics

Example processes from input

Common elements

Pattern 2:

Name and description

Defining characteristics

Example processes from input

Common elements

...

There are reasoning processes:

724

{extracted_responses}

Group similar reasoning processes into exactly 5 patterns. Please provide a classification of reasoning patterns, with each group defined by their common characteristics and examples from these processes.

725

726 D.2 Full Prompt for Responses Classification

Full Prompt for Responses Classification

Please classify each reasoning process into one of the following five patterns. Output the classifications in JSON format.

{patterns_categories}

Here are the reasoning processes to classify:

{extracted_responses}

Please output your classification in this JSON format:

727

728 D.3 Prompt Examples for Evaluation Tasks

Prompt Example for Easy Countdown

<lim_start>system

You are a helpful assistant. You first thinks about the reasoning process in the mind and then provides the user with the answer.<lim_end>

<lim_start>

user Using the numbers [1,2,3,7], create an equation that equals 5. You can only use basic arithmetic addition (+) and subtraction (-), and each number from the list can be used at most once. Show your work in <think> </think> tags. And return the final answer in <answer> </answer> tags, for example, to get 5 from [1,2,3,7], we have <answer> 2 + 3 = 5 </answer>; to get 13 from [1,2,3,7], we have <answer> 1 + 2 + 3 + 7 = 13 </answer>.<lim_end>

<lim_start>assistant

Let me solve this step by step.

<think>

729

Prompt Example for Long Arithmetic

<lim_start>system

You are a helpful assistant. You first thinks about the reasoning process in the mind and then provides the user with the answer.<lim_end>

<lim_start>user

Considering the experssion (((((((((16 + 47) + 92) + 84) + 5) + 49) + 82) - 33) - 99) - 69), you need to start from the innermost bracket and work your way out. You need to give the specific calculation process. You have to do a certain amount of calculations on each line, and print each calculation in the format of a + b = c or a - b = c, for example "1 + 2 = 3".<lim_end>

<lim_start>assistant

Let me solve this step by step.

<think>

730

Prompt Example for Number Theory

<lim_start>system

You are a helpful assistant.<lim_end>

<lim_start>user

You have four bags of copper coins. Each bag has the same number of copper coins. One day, you find a bag of 23 coins. You decide to redistribute the number of coins you have

731

so that all five bags you hold have the same number of coins. You successfully manage to redistribute all the coins, and you also note that you have more than 120 coins. What is the smallest number of coins you could have had before finding the bag of 23 coins?
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

732

Prompt Example for Geometry

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
Two sides of scalene $\triangle ABC$ measure 4 centimeters and 7 centimeters. How many different whole centimeter lengths are possible for the third side?
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

733

Prompt Example for Algebra

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
Six aluminum cans can be recycled to make a new can. How many new cans can eventually be made from 200 aluminum cans? (Remember that the first new cans that are made can then be recycled into even newer cans!)
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

734

Prompt Example for Calculus

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
For how many values of x in $[0, 2\pi]$ is $\cos^{-1}(\cos 4x) = \sin^{-1}(\sin x)$?
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

735

Prompt Example for Counting and Probability

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
7 white balls and k black balls are placed into a bin. Two balls are drawn at random. The probability that one ball is white and the other is black is $\frac{35}{66}$. Find the smallest possible value of k .
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

736

Prompt Example for AMC23-Q19

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
How many digits are in the base-ten representation of $8^5 \cdot 5^{10} \cdot 15^5$?
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

737

Prompt Example for AIME24-Q1

```
<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
Every morning Aya goes for a  $s$ -kilometer-long walk and stops at a coffee shop afterwards.
When she walks at a constant speed of  $s$  kilometers per hour, the walk takes her 4 hours,
including  $t$  minutes spent in the coffee shop. When she walks  $s+2$  kilometers per hour,
the walk takes her 2 hours and 24 minutes, including  $t$  minutes spent in the coffee shop.
Suppose Aya walks at  $s+\frac{1}{2}$  kilometers per hour. Find the number of minutes the
walk takes her, including the  $t$  minutes spent in the coffee shop.
Please reason step by step, and put your final answer within  $\boxed{\phantom{000}}$ .<lim_end>
<lim_start>assistant
```

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Prompt Example for OlympiadBench-Q1631

```
<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
For a positive integer  $a$ , define a sequence of integers  $x_1, x_2, \dots$  by letting
 $x_1 = a$  and  $x_{n+1} = 2x_n + 1$  for  $n \geq 1$ . Let  $y_n = 2^{x_n} - 1$ . Determine
the largest possible  $k$  such that, for some positive integer  $a$ , the numbers  $y_1, \dots, y_k$ 
are all prime.
Please reason step by step, and put your final answer within  $\boxed{\phantom{000}}$ .<lim_end>
<lim_start>assistant
```

739